has helped maintain the relevance of C++. This rule means that unnecessary features or operations do not affect the performance of the program. In C++, initializing a stack variable can be costly and can slow down the execution speed, resulting in variables containing indeterminate values. It is highly recommended to initialize primitive data type variables before using them.

8. Constants and Literals in C++

Constants are variables or values in C++ that cannot be modified once they are defined. They represent fixed values in a program and can be of any type, such as integer, float, octal, hexadecimal, or character constants. Each constant has a specific range. For instance, integers that are too big to fit into an int will be taken as long. The range of an int varies from -128 to +127 under the signed bit, and from 0 to 255 under the unsigned bit. In C++, literals and constants are used interchangeably. It is important to note that a constant's value remains the same throughout the program's execution, making it a useful tool for code optimization.

9. Operators in C++

Operators play a crucial role in the C/C++ programming language, as they allow programmers to perform various mathematical and logical computations on operands. Essentially, an operator is a symbol that operates on one or more operands to produce a result.

10. Loops in C+

In programming, loops are used when we need to execute a block of statements repeatedly. For instance, if we want to print "Hello World" ten times, we can achieve this by using either the iterative method or loops.

11. Access Modifiers

Access modifiers or access specifiers are an important feature of Object-Oriented Programming that enables data hiding. They are used in a class to define the accessibility of its members. Access modifiers impose restrictions on class members to prevent them from being directly accessed by external functions.

12. Storage Classes

In C++, Storage Classes are used to define the properties of a variable or function. These properties include the scope, visibility, and lifetime, which assist in tracking the existence of a variable or function throughout the program's runtime.